

Environment Setup Guide

Project: Long term health condition prevalence from 2021 census data

Document Version: 1.0.0 | **Target Stack:** Python 3.10

This document provides standardized, step-by-step instructions for establishing a fully reproducible execution environment for the project. Adhering to these specifications guarantees zero environmental divergence across local workstations, staging environments, and production clusters.

1. Prerequisites & System Requirements

Before proceeding with environment configuration, ensure your host machine satisfies the following hardware and software baselines:

Component	Minimum Specification
Operating System	Ubuntu 22.04 LTS / macOS Ventura+ / Windows 10 (WSL2)
Memory (RAM)	4 GB
Storage	Any
Package Manager	Anaconda (Python 3.10)

2. Option A: Local Development Setup (Conda & Pip)

Use this workflow if you are actively developing code, performing exploratory data analysis, or running local training loops.

Step 2.1: Clone the Repository

Clone the project repository from the corporate version control system and navigate into the root directory:

```
git clone https://github.com/Simbarashe-Mutyambizi/Data-Science-practise-PRT661.git
```

Step 2.2: Create and Activate the Conda Environment

We provide a locked environment configuration file `environment.yml` that pins exact library versions down to their build hashes.

```
# Create environment from specification file
conda env create -f environment.yml

# Activate the newly created environment
conda activate lthc-env
```

Step 2.3: Verify Package Integrity

Confirm that core dependencies such as PyTorch, Pandas, and Scikit-Learn resolve correctly:

```
pip install -r requirements.txt
```